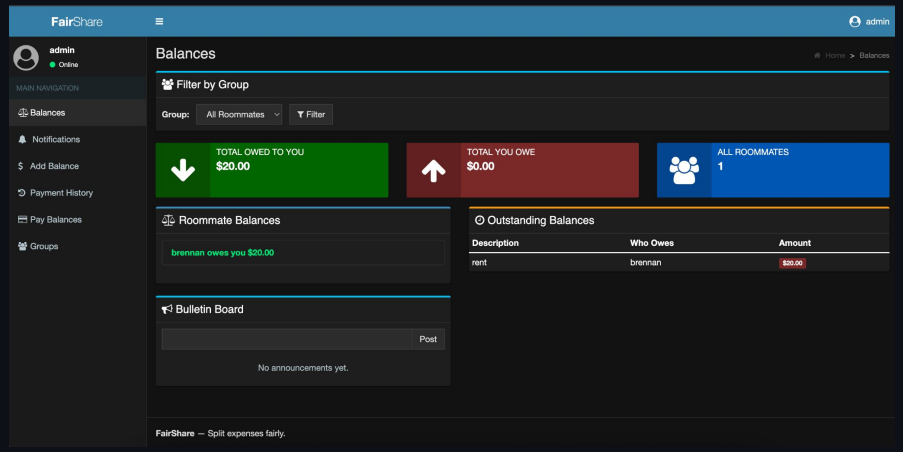
FairShare

Roommate Expense & Chore Management

CSCI 3308 — Group Project



Team Members

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PROJECT OVERVIEW

The Problem

Living with roommates creates recurring friction around shared expenses, chore assignments, and household communication. Spreadsheets are messy, group chats get buried, and informal IOUs are forgotten.

Split Expenses

Track who paid, who owes, and auto-calculate per-person shares across your household.

Manage Chores

Assign rotating chores to roommates and add a payment to those chores

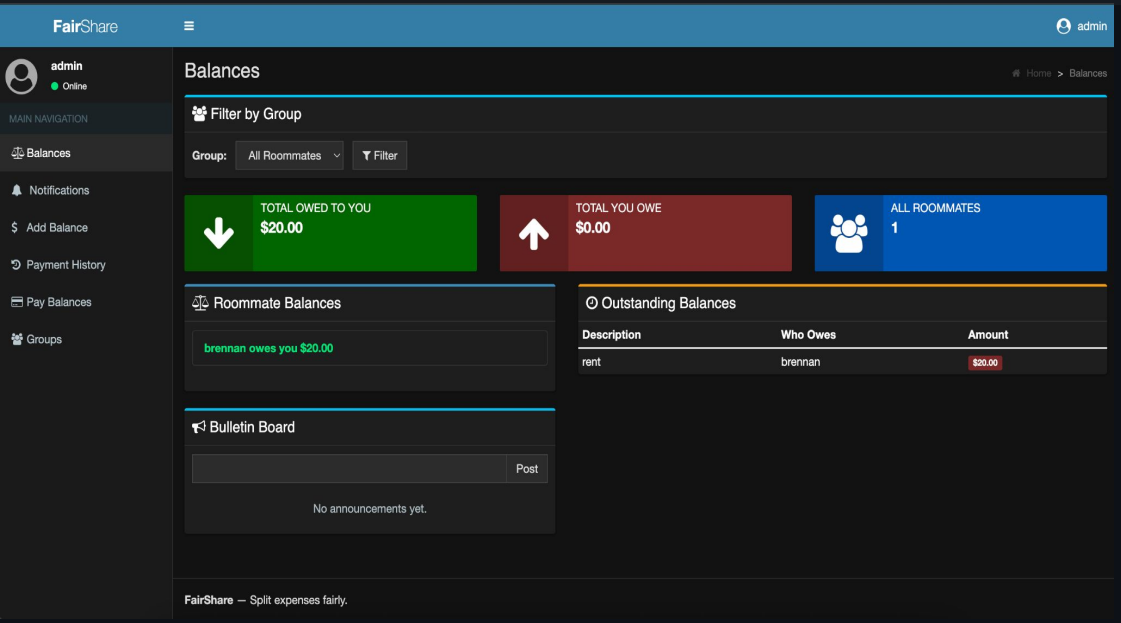
Group Invites

Token-based email invite system to securely onboard new roommates to existing groups.

Bulletin Board

Post announcements visible to all roommates — house rules, reminders, events.

KEY FEATURES



- Real-time balance tracking per roommate
- Mark expenses paid / outstanding
- Bulletin board for announcements
- Chore assignment & completion tracking
- Email-based group invites and expense notifications
- Full dark mode with OS preference sync

AUTOMATED TESTING

Mocha + Chai Automated testing suite

PASS	**[Positive] Server** GET /welcome → 200 + success message
PASS	**[Positive] Register API** POST /register with valid data → 200 + 'Success'
PASS	**[Negative] Register API** POST /register with empty fields → 400 + error message
PASS	**[Positive] Balances (Extra Credit)** GET /balances renders HTML page with 'Roommate Balances'
PASS	**[Negative] Balances (Extra Credit)** POST /mark-paid/not-a-number → 400 bad request

5 passing · 0 failing · All tests run via npm test (Mocha)

TOOLS & TECHNOLOGIES

Tool / Technology	Purpose	Category	Rating
GitHub	Version control & collaboration	VCS	★★★★★
GitHub Projects	Agile sprint & user story tracking	Project Tracker	★★★★☆
PostgreSQL 14	Relational database storing all app data	Database	★★★★☆
VS Code	Primary IDE for all team members	IDE	★★★★★
Node.js	JavaScript runtime / application server	App Server	★★★★★
Express 5	Web framework routing HTTP requests	Framework	★★★★★
Handlebars (HBS)	Server-side HTML templating engine	UI Tool	★★★☆☆
AdminLTE / Bootstrap	Responsive CSS UI component library	UI Tool	★★★★☆
Docker Compose	Containerize & orchestrate web + db services	DevOps	★★★★★

TOOLS & METHODOLOGIES

Tool / Technology	Purpose	Category	Rating
Render	Cloud deployment environment (live hosting)	Deployment	★★★★☆
Mocha + Chai	Unit & integration test framework	Testing	★★★★☆
bcryptjs	Password hashing before DB storage	Security	★★★★★
nodemailer	Send email invites for group onboarding	Additional	★★★☆☆
pg-promise	PostgreSQL ORM / query interface	Database	★★★★☆
Figma / Wireframes	UI wireframe design & planning	UI Tool	★★★☆☆

METHODOLOGIES

- **Agile / Iterative**
  Weekly sprint cycles with TA check-ins and continuous delivery
- **Feature Branching**
  Separate Git branch per user story; PR review before merging to main
- **Pair Programming**
  Complex features (auth, expense logic) built collaboratively
- **Peer Code Reviews**
  At least one teammate reviewed every pull request before merge

DATABASE DESIGN

PostgreSQL · 10 Tables

users

user_id, full_name, email, password

users_to_groups

user_id (FK), group_id (FK)

group_members

id, group_id, user_id

expenses

expense_id, description, amount, paid_by, group_id, category

balance_requests

request_id, requester_id, target_user_id, amount, status

roommate_groups

group_id, group_name

groups

group_id, group_name, created_by, created_at

group_invites

token (PK), group_id, email, created_at

expense_participants

participant_id, expense_id, user_id, amount_owed, is_paid

chores

chore_id, description, assigned_to, is_completed, completed_by

API Integration

The Engine

Fairshare moves beyond a static database by implementing an active notification system. Leveraging nodemailer and custom RESTful endpoints, we ensure users stay informed without needing to login and check the dashboard.

New Expense Alerts

Automated emails dispatched via automated mailer when a new expense is logged to reduce friction between roommates

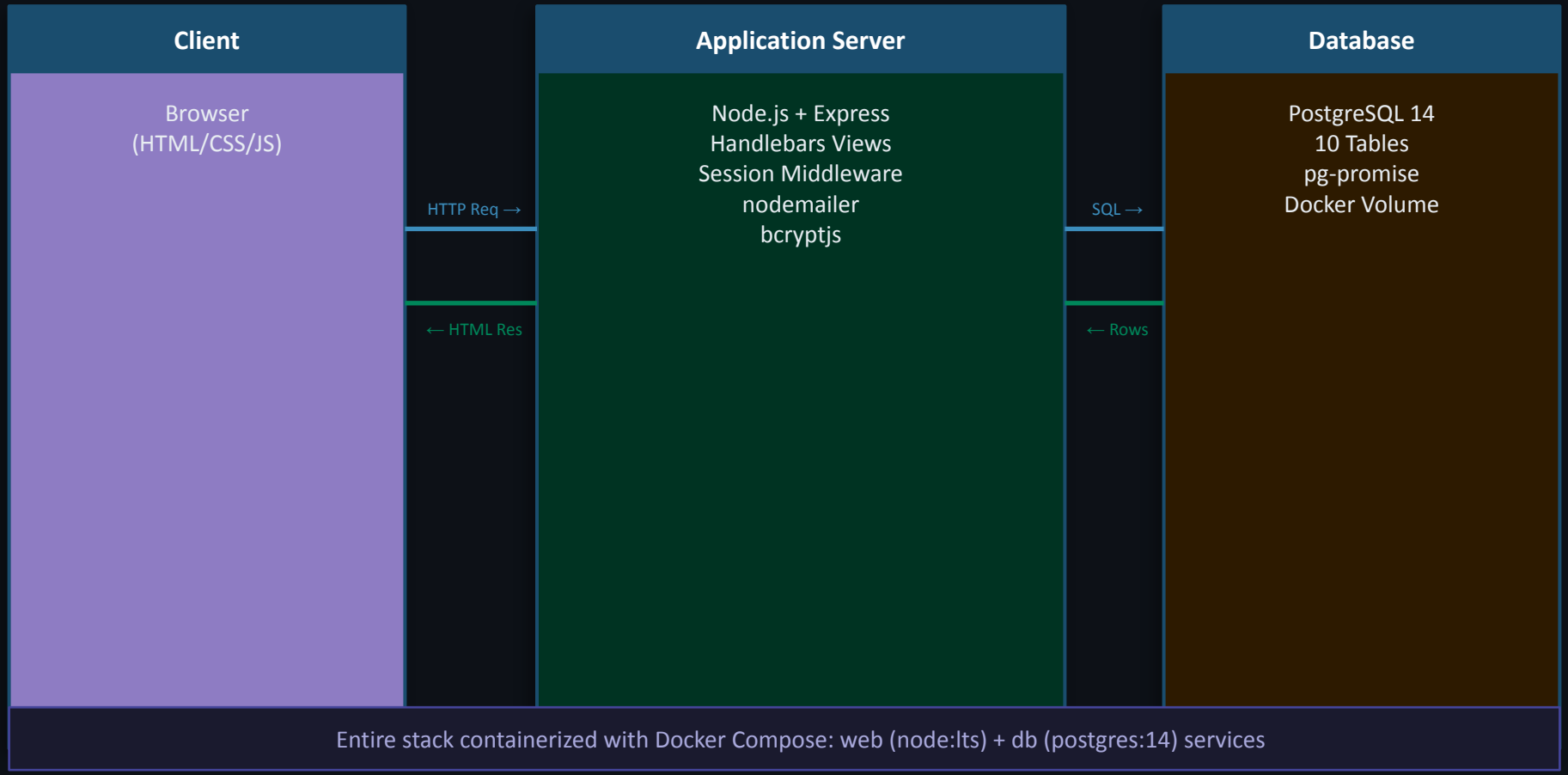
Secure Email Group Invites

Token based invitation system sent via Nodemailer to allow users to securely join specific groups

Internal Rest API

Bridge between our database and frontend, designed custom endpoints to interpret and process requests before sending emails

ARCHITECTURE DIAGRAM



CHALLENGES & SOLUTIONS

⚠️ Challenge

Roommate Balance Dashboard

✓ Solution

Regardless of the group you were in, it showed all users from the DB's balance. We fixed this by showing only users in your group while unrelated database users are filtered out.

⚠️ Challenge

Group Merge Conflicts

✓ Solution

We often were editing the same file or fixing the same issue and would commit these changing creating a merge conflict, to fix this we manually altered the code so the merge conflict is resolved.

⚠️ Challenge

Expense split precision

✓ Solution

JavaScript math like 60 / 3 can return 19.999999 instead of 20. Fixed by letting PostgreSQL handle all money math using its exact NUMERIC(10,2) type.

FUTURE SCOPE

What's next for FairShare?

01

Mobile App

iOS/Android app with push notifications for expense reminders, balance alerts, and chore due dates.

02

Venmo / PayPal Integration

One-click payment directly through linked payment processors, settling balances without leaving FairShare.

03

Recurring Expense Templates

Auto-create monthly bills (rent, utilities, internet) on a schedule so nothing gets forgotten.

04

Smart Chore Rotation

AI-assisted chore scheduling that balances workload fairly based on past history and member preferences.

05

Expense Analytics Dashboard

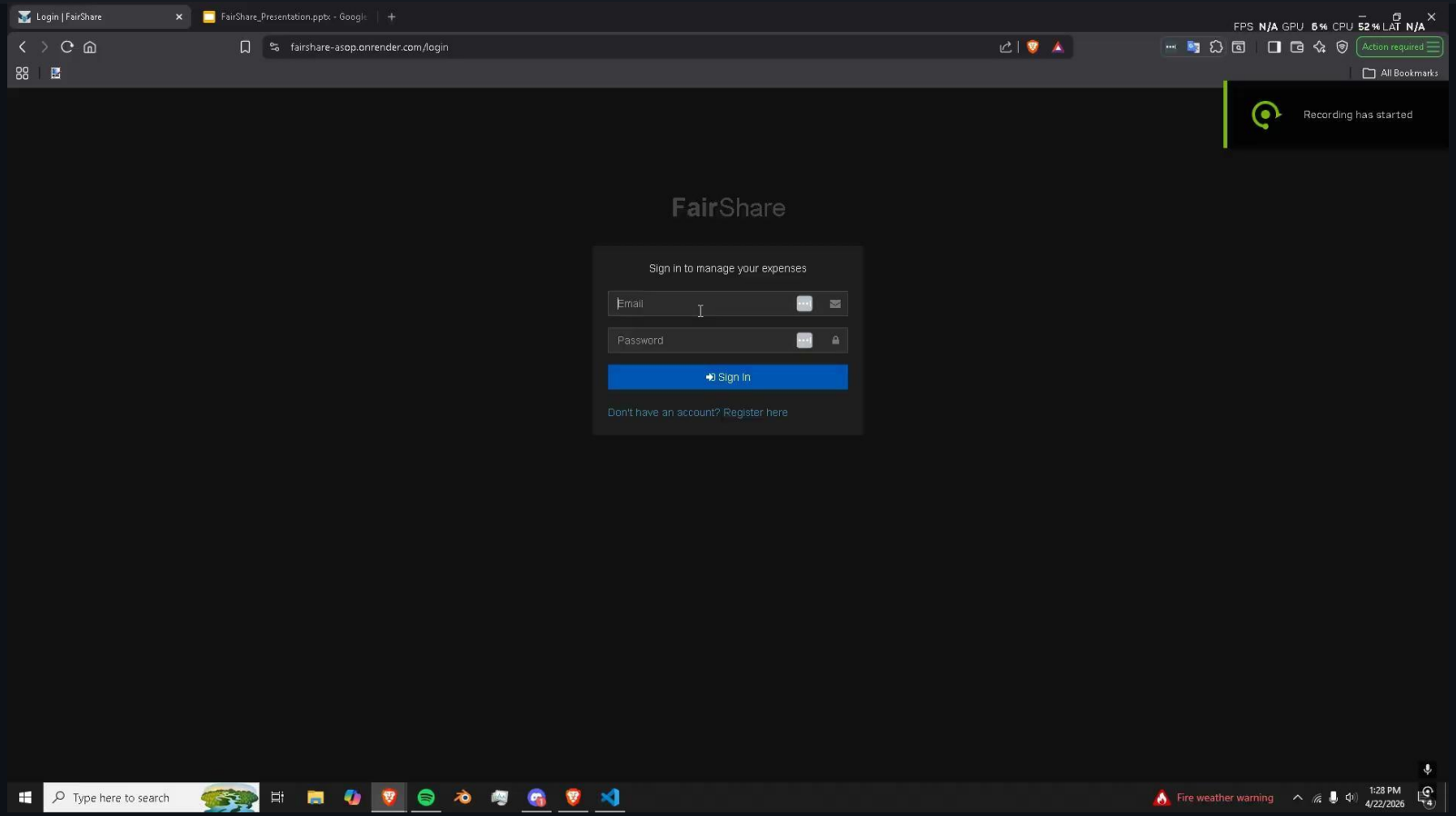
Monthly spending breakdowns with charts, by category, by person, by time period.

06

Multi-Group Support

Single user account spanning multiple living arrangements (college house + internship apartment).

DEMO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Don't have an account? Register here
Recording has started
Type here to search
Fire weather warning
1:28 PM
4/22/2026

FairShare

Making roommate life a little more fair.

Questions?